

N100 Financial Intelligence Platform

Analyst User Guide

Financial analytics, valuation, risk intelligence, cash-flow analysis, stock screening, peer comparison, sector analysis and automated reporting.

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Author: Tamilarasan M
Data Analyst | Data Science Enthusiast

1. Platform Overview

The N100 Financial Intelligence Platform is a financial analytics platform designed to analyze a large-cap company universe using financial statements, ratios, valuation, risk indicators, cash-flow metrics, peer comparison, sector analysis and automated reporting.

Core capabilities

- Financial ratio analysis
- Company financial profile analysis
- Stock screening
- Peer comparison
- Sector analysis
- Trend analysis
- Risk scoring
- Cash-flow intelligence
- Capital allocation analysis
- Valuation analysis
- Automated company tearsheets
- Automated sector reports
- Portfolio intelligence

Platform architecture

The platform uses a Streamlit dashboard as the interactive analyst interface. Streamlit communicates with a FastAPI backend through HTTP requests. The backend accesses the SQLite database and returns financial intelligence data.

Architecture: Streamlit Dashboard → FastAPI Backend → SQLite Database

Technology stack

- Python
- Pandas and NumPy
- Streamlit
- FastAPI and Uvicorn
- Plotly and Matplotlib
- Scikit-learn
- SQLite
- OpenPyXL
- ReportLab
- Pytest
- Requests

2. Installation and Application Startup

Create virtual environment

```
python -m venv .venv
```

Activate on Windows PowerShell

```
.venv\Scripts\Activate.ps1
```

Install dependencies

```
pip install -r requirements.txt
```

Start FastAPI backend

- Open Terminal 1.
- Activate the virtual environment.
- Run: `uvicorn src.api.main:app --reload`
- Backend URL: `http://127.0.0.1:8000`
- API base: `http://127.0.0.1:8000/api/v1`

Start Streamlit dashboard

- Open Terminal 2.
- Activate the virtual environment.
- Run: `streamlit run src/dashboard/app.py`
- Dashboard URL: `http://localhost:8501`

Important

Both FastAPI and Streamlit should be running simultaneously for the complete dashboard workflow.

3. Home Dashboard and Company Profile

Home Dashboard

The Home Dashboard provides a high-level overview of the company universe and key financial indicators.

- Total Companies
- Average ROE
- Average P/E
- Average D/E
- Revenue CAGR
- Debt-Free Companies
- Sector Distribution

Company Profile

The Company Profile page is used for detailed company-level financial analysis.

- Company information
- Financial ratios
- Profit & Loss
- Balance Sheet
- Cash Flow
- Revenue trends
- ROE versus ROCE trends
- Generated Pros & Cons

Suggested analyst workflow

- Select a company.
- Review the company information.
- Check profitability ratios.
- Review leverage and debt indicators.
- Review cash-flow trends.
- Review historical growth.
- Read the generated Pros & Cons.

4. Stock Screener

The Stock Screener allows analysts to filter companies using financial metrics and predefined screening conditions.

Screener features

- Custom metric filters
- Preset filters
- Financial screening
- Live results
- CSV download

Typical screening workflow

1. Open the Stock Screener page.
2. Select the required financial metrics.
3. Enter or select the desired filter conditions.
4. Apply the filters.
5. Review the resulting companies.
6. Download the results when required.

Analyst use cases

- Finding companies with stronger profitability
- Filtering companies by leverage
- Identifying companies using multiple financial conditions
- Creating a shortlist for deeper company analysis

5. Peer Comparison and Trend Analysis

Peer Comparison

The Peer Comparison page allows analysts to compare company KPIs with sector-level information.

- KPI comparison
- Sector average
- Company metrics

Recommended peer-analysis workflow

- Select the target company.
- Review its key KPIs.
- Compare the company metrics with the sector average.
- Identify relative strengths and weaknesses.
- Use the result as supporting evidence for further analysis.

Trend Analysis

- Multi-metric trend charts
- Growth percentage labels
- Historical financial trends

Trend analysis should be used to understand whether a company's financial performance is improving, stable or deteriorating over time rather than relying only on a single-year value.

6. Sector Analysis and Capital Allocation

Sector Analysis

- Sector bubble chart
- Sector KPI analysis
- Sector comparison

Available sectors

- Communication Services
- Consumer Discretionary
- Consumer Staples
- Energy
- Financials
- Healthcare
- Industrials
- Information Technology
- Materials
- Real Estate

Capital Allocation Map

- Capital Allocation Treemap
- Company distribution by allocation pattern

The Capital Allocation page helps analysts understand how companies are distributed across capital allocation patterns.

7. Automated Financial Reporting

Company Tearsheets

The platform automatically generates a financial PDF tearsheet for each analyzed company.

Metric	Result
Total Companies	92
Generated Reports	92
Skipped Reports	0
Failed Reports	0

Company reports are stored under `reports/tearsheets/`.

Sector Reports

Automated PDF reports are generated for the available sectors.

Current output: 10 sector reports.

Sector reports are stored under `reports/sector/`.

Portfolio Intelligence

- Total Companies
- Average Risk Score
- Average CFO Quality
- Average FCF Conversion
- Risk Distribution
- Highest Risk Company
- Lowest Risk Company

8. FastAPI Backend

The FastAPI backend provides REST API endpoints for serving financial intelligence data to the Streamlit dashboard.

Base URL

`http://127.0.0.1:8000/api/v1`

Health endpoint

`GET /api/v1/health`

Health response

The health endpoint provides API status, database row counts, application uptime and API version.

```
{"status": "ok", "version": "1.0.0"}
```

Main API modules

- Health monitoring
- Company listing
- Company profiles
- Stock screening
- Sector analysis
- Peer groups
- Peer comparison
- Portfolio analysis
- Valuation
- Annual reports and documents

Dashboard integration

The Streamlit dashboard communicates with the backend using `src/dashboard/utils/api.py`. The API client provides functions for health checks, company listing, company profiles, screener requests, sector data and companies by sector.

9. Data Quality and Testing

Quality assurance coverage

- API endpoint validation
- Data validation
- Missing data handling
- Duplicate company detection
- Financial calculation validation
- Cash-flow validation
- CapEx intensity validation
- Risk score validation
- Pros & Cons validation
- Report generation
- Dashboard integration
- Multi-page navigation
- Screener testing
- Valuation testing
- CSV export testing
- Excel export testing
- ETL validation
- KPI validation

Data quality checks

- Duplicate company IDs
- Null company IDs
- Invalid foreign keys
- Invalid financial values
- Invalid stock prices
- Negative stock volume
- Invalid market capitalization
- Invalid P/E ratios

Automated test result

Test Metric	Result
Total tests	152
Passed	152
Failed	0
Warnings	2

Overall	PASS
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Run the complete test suite with: `pytest -v`

10. Performance and Troubleshooting

Documented performance results

Test	Average	Slowest	Result
Screener API	0.042 sec	0.052 sec	PASS
Company Profile API	0.016 sec	0.029 sec	PASS

Troubleshooting: API does not start

- Confirm the virtual environment is activated.
- Confirm dependencies are installed.
- Run `uvicorn src.api.main:app --reload`.
- Check that port 8000 is available.

Troubleshooting: Dashboard does not load

- Confirm FastAPI is running.
- Confirm Streamlit is running.
- Run `streamlit run src/dashboard/app.py`.
- Open `http://localhost:8501`.
- Check the terminal for Python errors.

Troubleshooting: API data is unavailable

- Check the FastAPI health endpoint.
- Confirm the database is accessible.
- Check the API client configuration.
- Review the FastAPI terminal output.

11. Recommended Analyst Workflow

Step	Analyst Action
1	Start FastAPI and Streamlit.
2	Open the Home Dashboard.
3	Review overall company and sector statistics.
4	Use the Stock Screener to create a shortlist.
5	Open individual Company Profiles.
6	Review profitability, leverage and cash-flow indicators.
7	Review historical trends.
8	Compare the company with peers.
9	Review the sector context.
10	Review risk and valuation information.
11	Read the generated Pros & Cons.
12	Open the relevant company or sector PDF report.

Important output locations

output/cashflow_intelligence.xlsx

output/risk_scores.xlsx

output/valuation_summary.xlsx

output/peer_comparison.xlsx

output/screener_output.xlsx

output/pros_cons_generated.csv

output/distress_alerts.csv

output/capital_allocation.csv

reports/sector/

reports/tearsheets/

reports/pytest_report.html

12. Quick Reference

Purpose	Command / Location
Start API	uvicorn src.api.main:app --reload
Start Dashboard	streamlit run src/dashboard/app.py
API	http://127.0.0.1:8000
API Base	http://127.0.0.1:8000/api/v1
Dashboard	http://localhost:8501
Run Tests	pytest -v
HTML Test Report	reports/pytest_report.html
Company Reports	reports/tearsheets/
Sector Reports	reports/sector/

Project Summary

The N100 Financial Intelligence Platform combines financial data processing, financial ratios, valuation, risk intelligence, cash-flow analysis, capital allocation, peer comparison, sector analysis, stock screening and automated financial reporting.

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